

Guide to Writing Lab Reports

(Written by Nikki Pallat with Jacob Ritz and Virginia Gali)

Abstract

Your abstract should be very short (no more than 5 sentences). The purpose of this section is to **summarize your report**. You should not say anything in this section that is not covered elsewhere in the report.

You should include the following in your abstract:

1. State the physical principle you are testing (the motivating question of the report)
2. Explain how you tested the principle in lab
3. State the main quantitative result (with uncertainty)
4. The conclusion you made from your data

Introduction

Your introduction should just be one short paragraph. Three sentences is sufficient, but depending on the report (and your personal writing style), it could be a little bit longer. **Do not spend too much time on this section.**

You should include the following three things in your introduction:

1. Set up the hypothetical scenario (see the “Before You Start” section of the lab)
2. State the question
3. Provide a brief statement of how you intend to answer the question (i.e. what you used in lab to model the hypothetical scenario)

Prediction

Your prediction is typically based on equations or graphs. You **always** have to justify your prediction with physics. Your data does not have to agree with your prediction, but the physics presented in this section has to be correct. The prediction and analysis sections are the two most important sections that contribute to your grade

You can also think of this section as the “theory” section. You should explain what you would expect to get according to what you have learned in lecture and from your textbook. For example, if you drop a ball, you would theoretically expect its acceleration to be 9.8 m/s^2 ; you would explain to the reader why you would expect that and where that value comes from.

You should include the following in your prediction section:

1. An equation or graph (when appropriate)

2. A proposed answer to the question of the report (the prediction)
3. Justification of prediction with physics

Procedure

This section should just be one paragraph. **This is the least important part of your lab report** and should be pretty simple to write. Do not include obvious or unnecessary steps. Make sure to avoid using first or second person.

You should include the following in your procedure section:

1. Provide a reasonable amount of detail where the reader could *roughly* replicate the experiment
2. Explain how you determined the uncertainty of measurements
3. Mention any systematic errors that may have caused you to get results that disagree with the expected result (you will discuss these errors in more detail later, just mention them here)

Analysis

This is the most important section and where most of your grade comes from!!

In this section, you should accept or reject the prediction you made earlier in the report. Use quantitative analysis, avoid vague terms (such as “very close to”). **Split your analysis into three short paragraphs.** You may rearrange these paragraphs based on your own personal writing style, but please keep them separate.

Paragraph 1: Results and Uncertainties

In the first paragraph, **present the result** of your data and **show how the prediction is incorrect or correct**. Then **discuss the uncertainty** of your measurements, and discuss if this does or does not change your result. This must be **quantitative**. See the examples below to understand what I expect by quantitative uncertainty analysis (these are not examples of the entire paragraph).

Example 1: “Since the uncertainty for the acceleration measurements was $\pm 0.2 \text{ m/s}^2$, the measured accelerations of the tennis ball (9.7 m/s^2) and the lacrosse ball (9.8 m/s^2) are not statistically significantly different. Therefore, our data suggests the acceleration is constant regardless of mass.”

Example 2: “As shown in Figure 2, the speed of cart A was $5.2 \pm 0.1 \text{ m/s}$, and the speed of cart B was $2.1 \pm 0.1 \text{ m/s}$. When considering the uncertainty of our measurements, the minimum actual speed of cart A could be 5.1 m/s and the maximum actual speed of cart B

could be 2.2 m/s. Therefore, our data shows the speed of cart A is certainly greater than the speed of cart B.”

Paragraph 2: Errors

In the second paragraph, discuss the **potential sources of error** in your experiment and how these errors likely did or did not affect the conclusion of your report. **Never say “human error”** because this is vague and meaningless to the reader and does not demonstrate any understanding of the physics happening. Do not just list sources of error here. Make sure you **explain (1) what the source of error was** and **(2) how it could have affected your data. If it is systematic you must do a calculation to show how it would have biased your results!** Notice that some types of error will affect your results consistently and some affect your results unpredictably. Note that random errors are completely random (unpredictable) and cannot be replicated by repeating the experiment again. Meanwhile, systematic errors produce consistent errors that shift your results by a fixed amount.

Example 1 (systematic error): “One potential source of error is that the initial calibration of our video in Vernier could have been incorrect. The average acceleration of the ball was 10.9 m/s^2 rather than the expected acceleration due to gravity (9.8 m/s^2). If the meter stick in the video was incorrectly calibrated as 10% smaller than 1m, all of the measured accelerations would have increased by this factor.”

Example 2 (random error): “A potential source of error could have been incorrectly tracking the block in Vernier. Through an inconsistent spot of tracking, the recorded distance traveled for each time interval may have been shifted lower or higher than the actual distance traveled. The error in the distance measurement would have, likewise, shifted the calculated acceleration.”

Paragraph 3

In the third paragraph, **answer the report’s question** (posed in the introduction) and **justify it with physics**. Explain to the reader what physics is happening in the experiment. Connect these physical concepts to potential applications or the overarching concepts of the chapter. In other words, answer the question: “So what? Why do we care?” Additionally, if your prediction was incorrect, discuss potential factors we did not consider in the prediction. An example of this would be if we did not consider air resistance but that potentially impacted our results.

Graphs and Tables

Throughout this section, you should include relevant, well-constructed graphs and figures. **Do not dump raw data**; only include what is helpful and relevant to the reader. Remember that you must

include **units and uncertainties for all measurements** (even time because that is a measurement too!), which means you must **include error bars on all plots**. Make sure to include labels (i.e. “Figure 1”) and descriptive captions for all plots and tables. Use these labels to refer to the figures in your analysis.

The plots in this section should include:

1. Include axes labels (with units!) and fit equations on plots
2. Uncertainties and units for all measurements (including error bars on all plots)
3. Reasonable uncertainty values
4. Descriptive captions and labels
 - Every plot needs a caption that says more than “a plot of position v.s. Time.”
 - Use this space to briefly discuss the plot: “Vernier pro data plot as position v.s. Time. The slope of the line of best fit was 10.2 ± 0.1 , meaning the cart had an average velocity of 10.2 ± 0.1 m/s”

Conclusion

This section should just be one paragraph wrapping up your report. You should not be saying too many new things in this section.

You should include the following (likely in this order) in your conclusion:

1. A brief summary of what you did and the question/purpose of this report
2. What your result was
3. Compare your result to your prediction
4. Interpret your result in the context of hypothetical scenario

This should be quite similar to your abstract, but in your abstract you assume your reader has not read your paper, but in the conclusions you assume they have.

General Formatting Notes

- Your report should be 4 pages maximum. If you can write a great report in 3 pages please do so!
- Your report should be single-spaced or double-spaced (TA's preference)
- Graphics should be clear and easy to read titles, labels, error bars, and descriptive captions
- If you include an equation in your report, put it on its own line and make sure to define the variables and constants in the equation
- Keep tense consistent throughout

- Use scientific and technical language
- Keep writing concise
 - Example (Good): The error in position is 1mm since measurements are limited by the pixel size for choosing points in vernier pro.
 - Example (Bad): In order to access the error owing to positional uncertainty, we employ a simple assumption that the computer monitor pixel size is the minimum resolvable distance, making 1mm a sufficient choice for positional uncertainty.
- Avoid typos and informal language
- Limit use of first and second person (less formal)
- **Read your report all the way through at least once before turning it in! This helps you catch minor errors and helps make sure your arguments flow well**